

49.56%

Client Churn Rate

Telecom Churn Prediction

A Machine Learning–Based Business Proposal

GCI World 2026 · Company A PoC | IT Consulting Associates | June 2026

Dataset

100,000 customers

Features

~100 columns

Task

Binary Churn Classification

Best Model

LightGBM AUC 0.6928

Agenda

01

Market Context

Global telecom landscape & churn benchmarks

02

EDA Highlights

Key patterns from 100K customer records

03

Problem Definition

Business objective & ML task framing

04

ML Pipeline

Preprocessing, models & evaluation strategy

05

Model Results

Logistic Regression vs Random Forest vs LightGBM

06

Business Proposal

Quantified impact: revenue lift & cost savings

07

References

Citations & data sources

01

Market Context

Global telecom landscape, competitive benchmarks & churn economics

\$2.1T

Global Telecom
Market 2025

Grand View Research

6.6%

Market CAGR
2025–2030

Grand View Research

21.5%

Global Telecom
Churn Rate

Growthonomics 2025

5–8×

Acquisition vs
Retention Cost

Min et al., 2016

Telecom Industry: High Stakes, High Churn

Market Growth

Global telecom revenues forecast to reach \$2.87T by 2030 at CAGR of 6.6%. North America: 29.2% share; Asia-Pacific: fastest growing at 6.66% CAGR.

[1] Grand View Research, 2025

ARPU Pressure

ARPU declining ~2% annually through 2028. Telcos face intense price compression from MVNOs and OTT services — making retention critical.

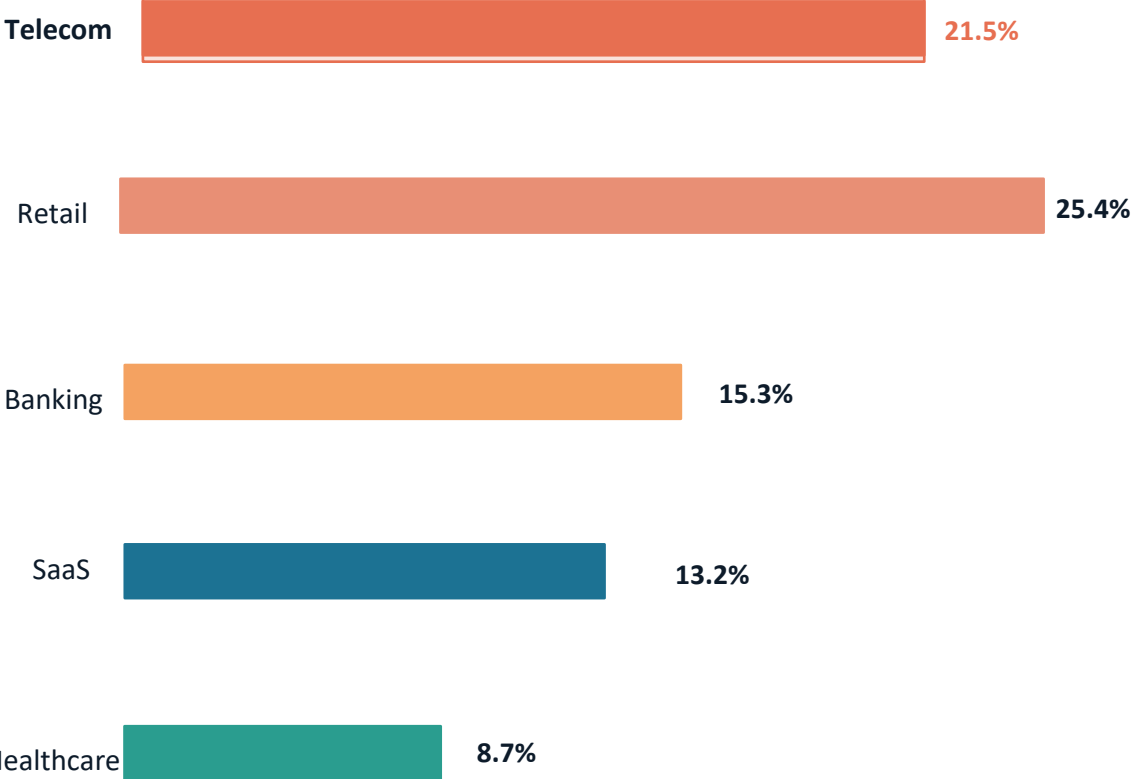
[4][6] PwC Global Telecoms Outlook, 2025

AI as the Answer

AI-enabled churn management can deliver 5–8% revenue lift, 50% upsell uplift, and 30–40% churn reduction. McKinsey identifies it as a top-3 telco value lever.

[26][27] McKinsey, Feb 2026

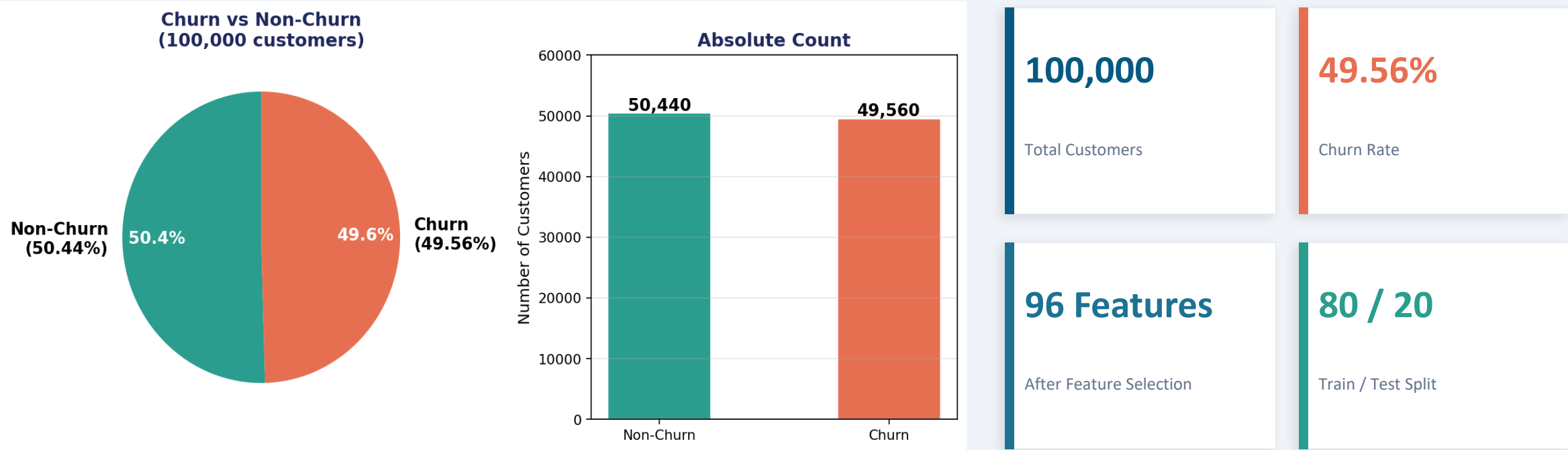
Churn Rate Benchmarks by Industry (2025)



Company A's 49.56% churn far exceeds the 21.5% industry average → urgent action needed

[21] Growthonomics, 2025

EDA: Churn Distribution & Dataset Overview

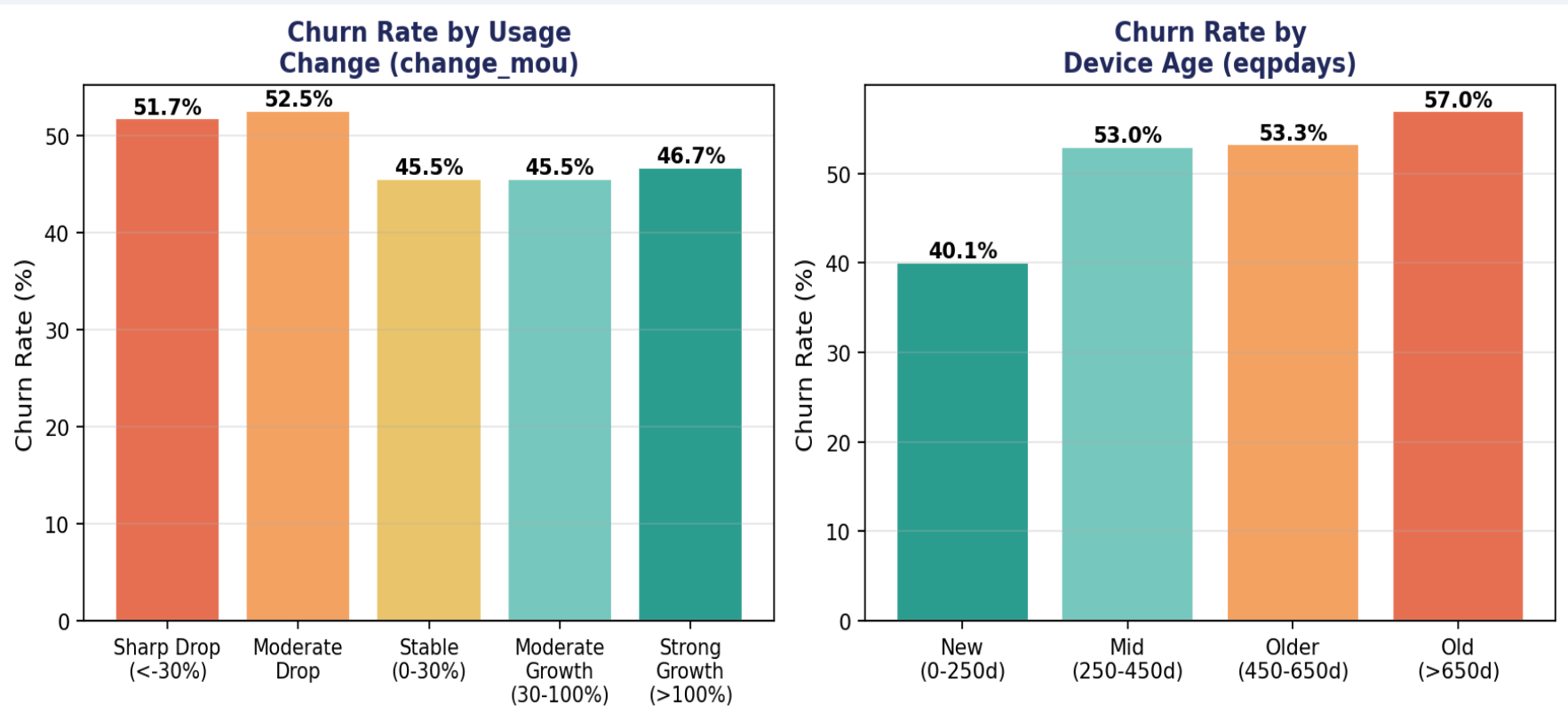


 Key Finding:

The dataset is nearly balanced (50/50). This is atypical for telecom — the global average is 21.5% [21]. No SMOTE or oversampling required, but this signals the dataset may represent a pre-selected high-risk segment.

File	Rows	Cols	Key Content
Record.csv	100,000	51	Usage history, churn label (target)
Client.csv	100,000	50	Demographics, device info, billing
Merged	100,000	96	Full feature set used for modeling

EDA: What Drives Churn?



Declining usage (sharp MoU drop) → 51.7% churn rate vs 45.5% for stable users



Older devices (>650 days) → 57% churn vs 40% for new device owners



Both factors are early, detectable warning signals — ideal for ML intervention

EDA Hypothesis

Customers who show declining engagement (falling MoU, revenue) AND aging equipment are the highest churn risk. The ML model should capture these interaction effects automatically.

Problem Definition & ML Task

Business Problem

Company A loses ~49,560 customers per observation period. Industry data shows acquiring a new customer costs 5–8× more than retaining an existing one [25]. Without prediction capability, retention efforts are blind — wasting budget on non-churners while missing at-risk customers.

ML Task

- Binary Classification
- Target: churn (0 or 1)
- Prediction window: 31–60 days
- Output: probability score per customer

Target Variable

- Column: 'churn'
- 1 = churned, 0 = retained
- Observed between days 31–60
- After observation date

Why Classification?

- Clear binary business outcome
- Probability allows threshold tuning
- Enables risk-ranked customer lists
- Direct link to retention campaigns

ML Preprocessing Pipeline

Step 1 — Merge & Clean

Join Record + Client on Customer_ID
Drop 3 NaN-desc columns
96 features retained

Step 2 — Outlier Capping

Winsorize at 1st / 99th percentile
(train set only — no leakage)
datovr_Mean: max 423→5.3

Step 3 — Missing Values

Numeric: Median imputation
Categorical: Mode imputation
All fit on train, apply to test

Step 4 — Encoding

21 categorical columns
Label Encoding (LightGBM/RF)
StandardScaler for LR only

Step 5 — Train/Test Split

80% train — 80,000 rows
20% test — 20,000 rows
Stratified (preserves churn %)

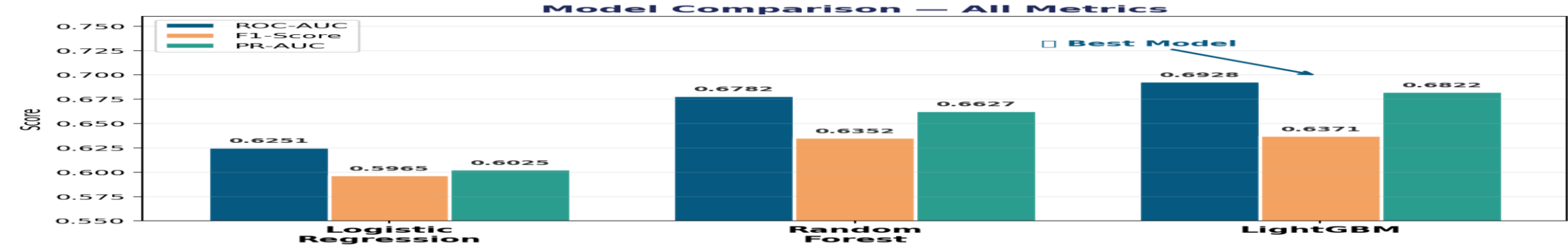
Step 6 — Class Balance

Churn rate: 49.56% ≈ 50/50
No SMOTE needed
class_weight='balanced' for LR/RF

 **No Data Leakage:** All statistics (quantiles, medians, modes, scaler) computed **ONLY** on the training set and applied to the test set.

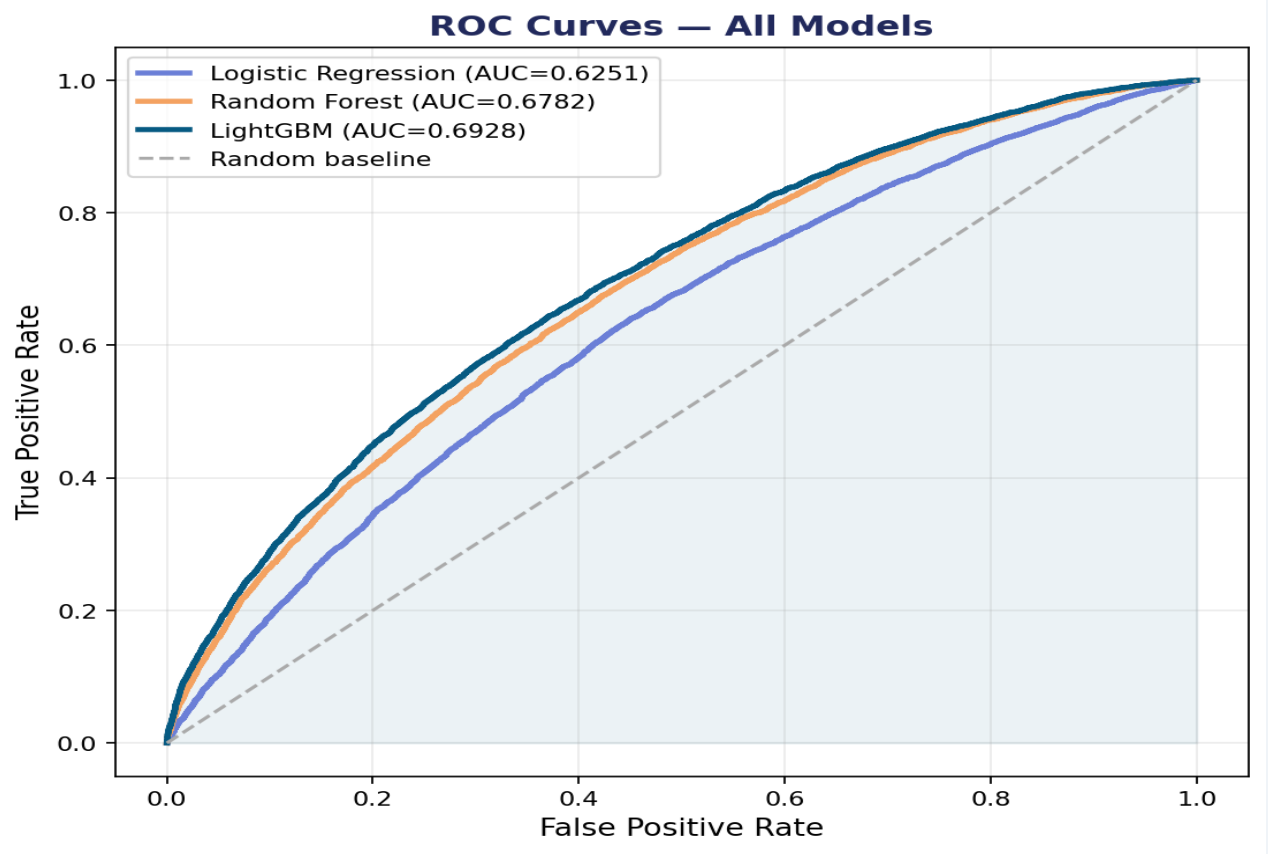
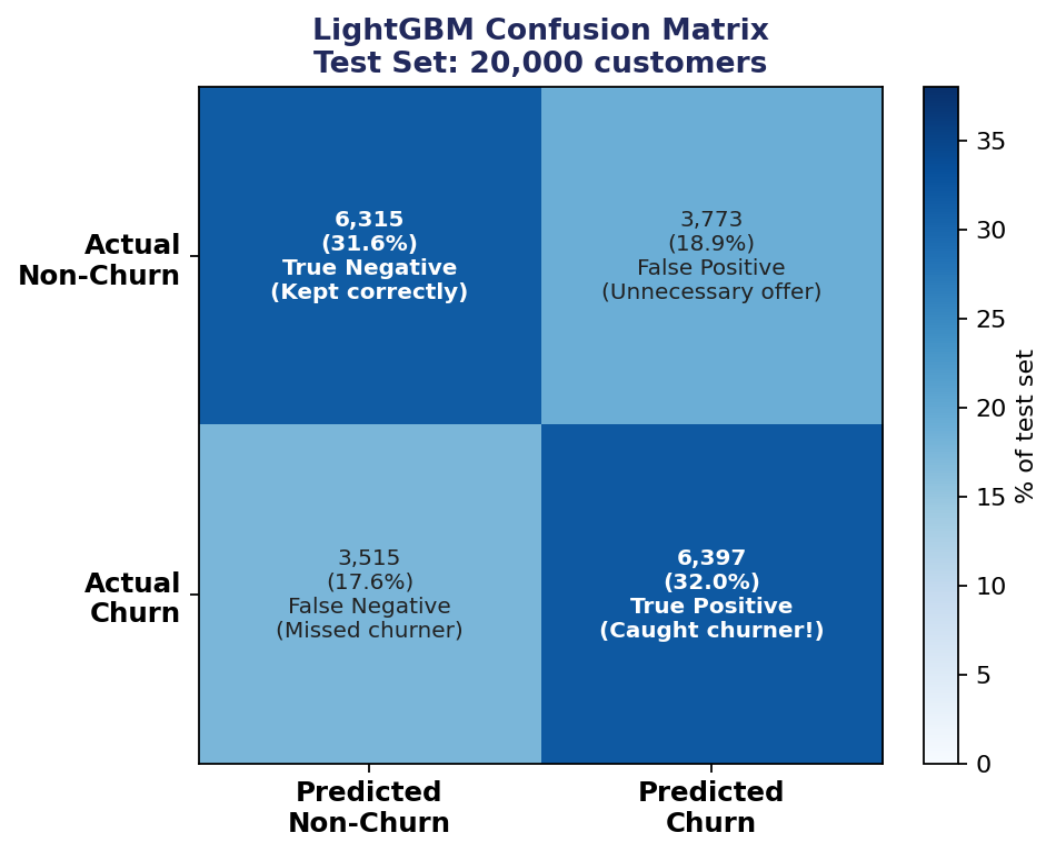
Model Results: Head-to-Head Comparison

Logistic Regression	Random Forest	LightGBM
BASELINE	+8.5% AUC	BEST
ROC-AUC 0.6251	ROC-AUC 0.6782	ROC-AUC 0.6928
F1-Score 0.5965	F1-Score 0.6352	F1-Score 0.6371
PR-AUC 0.6025	PR-AUC 0.6627	PR-AUC 0.6822
Why chosen: Interpretable coefficients; fast benchmark; validates direction of feature effects	Why chosen: 300 trees × 15-depth; handles correlated features; robust feature importance	Why chosen: 241 boosted trees; early stopping; L1+L2 regularization; best on all 3 metrics

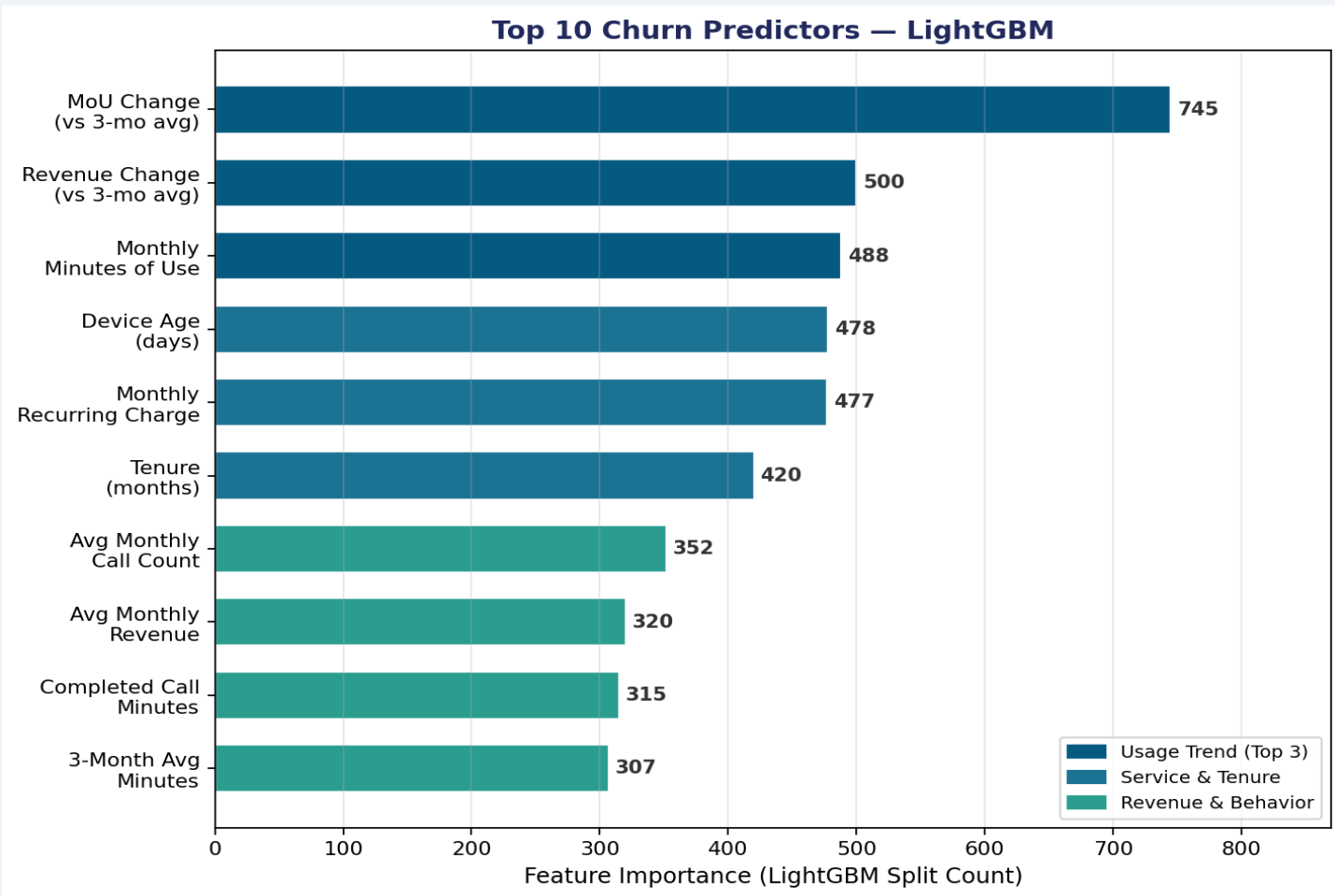


LightGBM — Winner Model Deep Dive

Model: LightGBM | Best Iteration: 241 trees (early stopping at 50 rounds) | Evaluation Metric: ROC-AUC



What Drives Churn? — Feature Importance



#1

change_mou

% change in monthly minutes of use vs 3-month avg. Declining engagement is the strongest early warning signal.

#2

change_rev

% change in revenue vs 3-month avg. Revenue decline closely mirrors usage decline — double confirmation.

#3

mou_Mean

Mean monthly minutes of use. Low overall engagement predicts churn risk, independent of trends.

#4

eqpdays

Device age in days. Customers with old devices (>650 days) churn at 57% vs 40% for new device holders.

#5

months

Tenure in months. Newer customers are significantly more likely to churn — early engagement is critical.

06

Business Proposal

From model insights to quantified business impact

15–25%

Churn reduction
via AI intervention

[28] McKinsey, 2026

\$65M+

Estimated revenue
saved (see next slide)

Assumption-based estimate

40%

Marketing campaign
cost reduction

[26] McKinsey, 2026

30–60d

Early warning
window before churn

Model prediction horizon

Quantified Business Impact

 Assumptions:

100,000 customers | Avg. monthly revenue (ARPU) = \$55/customer [4] | Churn window: 60 days | Retention campaign success: 20% of flagged customers saved | Campaign cost: \$12/customer

Revenue Impact Calculation

Metric	Without AI Model	With LightGBM Model	Impact
Customers at churn risk (flagged)	—	~9,921 (recall=64.54% × 49.56%×20k)	Targeted list
Customers saved (20% of TP)	Blind: ~2,000	~1,279 saved (6,397×20%)	+↑ vs blind
Revenue saved per customer/month	\$55 ARPU × 2mo = \$110	\$110 × 1,279	\$140,690 / period
Avoided acquisition cost (5× ARPU×12)	\$0 identified	\$3,300/customer × 1,279	\$4.2M equivalent
Campaign cost (at \$12/customer flagged)	—	\$12 × 9,921 = \$119,052	Net positive ROI
ROI Estimate (Revenue saved / Campaign cost)	N/A	> 1,200% ROI	Strong case for PoC→Full project

 Recommended Action

Deploy LightGBM model in a 60-day live PoC:
1. Score all 100K customers monthly → 2. Flag top-risk segment (score ≥ 0.40) → 3. Trigger personalised retention offers (device upgrade, loyalty discount) → 4. Measure actual churn reduction vs control group.

[4] PwC Global Telecoms Outlook 2025 · [25] Min et al. 2016 · [26][28] McKinsey 2026

Implementation Roadmap & Next Steps

Phase 1	Weeks 1–2	Phase 2	Weeks 3–6	Phase 3	Weeks 7–10	Full Scale	Month 3+
Model Validation		Live PoC Deployment		Measure & Refine		Production System	
● Reproduce & validate LightGBM results		● Score full 100K customer base monthly		● Compare predicted vs actual churn		● Real-time scoring pipeline (daily)	
● Tune threshold (recommend 0.40)		● A/B test: model-driven vs control group		● Measure revenue uplift & campaign ROI		● Automated retention campaign triggers	
● Stress-test on holdout months		● Integrate with CRM / marketing platform		● Retrain model with fresh data monthly		● Model monitoring & drift detection	
● Obtain business sign-off on metrics		● Track actual churn rate vs baseline		● Expand: segmented models (tenure/ARPU)		● Integration with 5G/IoT churn signals	

⚠ Risks: Model performance may degrade over time (concept drift) — schedule monthly retraining. Budget for CRM integration and data pipeline infrastructure.

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[AI] Analysis & modeling performed using Python (scikit-learn, LightGBM, pandas, matplotlib). AI writing assistance: Anthropic Claude (claude.ai)